

Abdul Hadi

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Objective

BS Artificial Intelligence student at FAST with hands-on experience in computer vision, embedded systems, and software development. Seeking an AI/ML internship to apply practical skills in real-world systems while gaining industry exposure.

Education

National University of Computer and Emerging Sciences (FAST NUCES)
BS in Artificial Intelligence

Islamabad
Aug 2024 – Present

Experience

Lab Demonstrator · Part-time

FAST-NUCES, Islamabad | Jan 2026 – May 2026

Conducted weekly C++ OOP lab sessions for first-year Cybersecurity students, covering core concepts like classes, inheritance, and polymorphism. Graded assignments and resolved student queries.

Projects

NeoPet – PETIFI

Python, C++, TensorFlow Lite, MediaPipe, OpenCV, MQTT, ESP32

- Achieved approximately 95% face recognition accuracy using MobileFaceNet (TensorFlow Lite) and OpenCV, triggering intruder alerts (buzzer + red LEDs) with a QR code verification layer within a 3-second response latency.
- Enabled hands-free navigation by mapping 4 MediaPipe hand gestures to real-time motor commands, relayed from host laptop to ESP32 via MQTT for hardware execution.
- Resolved Python 3.12, MediaPipe, and Protobuf conflicts and voltage-induced ESP32 damage through iterative circuit protection, achieving stable real-time AI performance on constrained hardware.

EduVault

Node.js, Express.js, MySQL, JavaScript, HTML/CSS, JWT, bcryptjs, Multer

- Built a full-stack MVC platform with RESTful APIs enforcing role-based access control across 3 roles, domain-restricted auth, roll number validation via regex, and stateless JWT sessions with bcrypt-hashed passwords.
- Reduced content access friction by building a resource pipeline with Multer for multi-format uploads, an admin moderation workflow for resources and teacher accounts, audit-logged downloads, and in-browser PDF preview.
- Achieved API response times of 61-118ms across all core endpoints by implementing live keyword search, hierarchical subject/topic categorization, bookmarking, and a 5-star rating system on a MySQL backend.

AEGIS

Python, ROS2, YOLOv8, MediaPipe, MQTT, ESP32, Ollama

- Built an autonomous emergency response robot that detects victims, fire, and rubble using YOLOv8 object detection, assesses victim condition via MediaPipe pose estimation, and generates a real-time 2D map of the area using ORB-SLAM.
- Implemented acoustic event triangulation to detect human screams and navigate them, alongside a flame sensor and HC-SR04 ultrasonic sensors for fire detection and obstacle avoidance.
- Integrated an LLM-based tactical voice system (Ollama/TinyLlama) running on a host machine for direct robot communication and status updates, relayed to ESP32 via MQTT.
- Engineered a path-finding module using A-Star algorithm to calculate the shortest evacuation route, with environmental monitoring for temperature and smoke levels via MQ-2 and DHT sensors.

Skills

- Programming Languages:** Python, C++, JavaScript, Arduino C, HTML, CSS
- Web & Backend:** Node.js, Express.js, REST APIs, MySQL, JWT, bcryptjs, Multer
- Libraries:** NumPy, Pandas, Matplotlib, Seaborn, TensorFlow Lite, OpenCV, MediaPipe, YOLOv8, Ollama
- Tools & Platforms:** Git, GitHub, VS Code, MySQL Workbench, Arduino IDE, Jupyter Notebook, ROS2, MQTT

Certifications

- Robotics & Hardware Systems** – FAST-NUCES, Department of AI & DS
- Innovating with Google Cloud Artificial Intelligence** – Google Cloud